
Physics-Posterior Distillation for Trustworthy Methane Source Attribution from Sparse Sensors

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Abstract

Rapid repair of methane super-emitters is among the highest-leverage near-term climate actions, and satellite and sensor programs now detect far more releases than are acted on: attributing a detection to a specific source, with *trustworthy* uncertainty, is the bottleneck. A fast neural localizer can output a source heatmap over a facility, and split conformal prediction can turn it into a region that provably contains the true source 90% of the time — but coverage alone does not say whether that region is *actionably small*. We study this gap on CH₄-500, a methane-configured testbed (500 m site, steady-state Gaussian plume with Pasquill–Gifford stability classes, 10–500 kg/h releases spanning the super-emitter threshold, ppm-level sensors) whose tractable physics lets us compute the exact Bayesian posterior of every scenario. These exact posteriors serve twice — as a dense training signal and as the reference that measures how close a learned localizer comes to the physical information limit. Standard location-label training yields a calibrated localizer whose 90% regions are, on identifiable scenarios, a median 32× larger than the data permit. Distilling mildly tempered exact posteriors into the *identical* network — changing only the training target — cuts this to 8.5×, sharper on 94% of scenarios (paired $p < 10^{-12}$), while barely changing point accuracy; raw untempered targets fail to optimize. A stratified audit shows the gap is model-limited only where sensors are dense, and that over half of random sparse layouts are physically un-localizable — a network-design finding. The simulator, exact posteriors, and full pipeline are released.

1 Introduction

Methane is responsible for roughly a third of observed warming to date, and because it clears the atmosphere in about a decade, cutting it slows warming on human timescales; oil-and-gas “super-emitters” — point releases of ≥ 100 kg/h — are a small, individually fixable share of the total, making their rapid repair one of the highest-leverage climate actions available. Detection is increasingly solved: the UNEP Methane Alert and Response System issued $\sim 1,200$ satellite notifications in two years, of which roughly 1% drew a documented response [UNEP, 2024]. A principal obstacle is *attribution*: turning a coarse detection into a trustworthy statement about *which* facility or equipment is leaking. Regulation now makes this concrete — under the U.S. Super-Emitter Program obligations attach to operators within 50 m of a reported location [U.S. EPA, 2024], and EU Regulation 2024/1787 mandates energy-sector methane MRV [European Union, 2024] — so an attribution’s *uncertainty* is itself operationally and legally material.

A learned localizer trained on dispersion simulations can map sparse sensor readings to a source heatmap in milliseconds, and split conformal prediction can wrap that heatmap in a region with a finite-sample coverage guarantee. This is necessary but not sufficient. Coverage is a statement about

frequency, not *efficiency*: a 90% region covering half the site is calibrated and useless, sending crews to search far more area than the readings justify. What one wants is a region as small as the physics permits — and, crucially, a way to know whether a given localizer achieves that.

Our observation is that in simulation-based training the exact Bayesian posterior of every scenario can be computed offline and used two ways at once: as the reference that measures how close a localizer comes to the physical information limit, and as a dense training signal that closes much of the gap. Standard training teaches only the realized source (“the leak was here”), never the shape of the uncertainty — which alternative locations fit, when the posterior is a ridge or multimodal, how much a sensor tightens it. The exact posterior teaches exactly that. We study this on a methane-configured testbed and find: a calibrated baseline whose regions are a median $32\times$ too large on identifiable scenarios; a physics-distilled variant — same network, same data, only the target changed — that cuts this to $8.5\times$ with point accuracy unchanged; a tempering requirement (raw posterior targets fail); and a stratified picture that turns the audit into sensor-network guidance.

Contributions. (1) *Physics-posterior distillation*: training an amortized localizer against exact simulator posteriors as soft targets improves calibrated-region sharpness substantially (identifiable-scenario inefficiency $32\times \rightarrow 8.5\times$, noise-response ratio $0.001 \rightarrow 0.72$) at no inference cost, with point accuracy nearly unchanged — isolating the deficiency as posterior approximation, not localization. Mild teacher tempering is a necessary ingredient. (2) *An exact-posterior audit*: conformal calibration equalizes coverage across models so region size becomes a pure sharpness measurement against the physical limit; a dose–response protocol tests whether uncertainty reacts to sensors, geometry, and noise as the physics dictates; stratification localizes the amortization gap. (3) *A released methane-configured benchmark* (CH4-500) shipping exact per-scenario posteriors — the enabling artifact for both the supervision and the measurement.

Scope. CH4-500 is a controlled testbed, not a deployment model: steady-state 2-D-sensed Gaussian plume, flat terrain, known stability, fixed release height. The idealization is deliberate — it is what makes the exact posterior computable, so the trust machinery can be validated against ground truth. The path to the field runs through recalibration on controlled-release data (e.g., METEC ranges); we claim the reliability layer, not field readiness.

2 The CH4-500 Testbed

Physics. A steady-state Gaussian plume on a 500×500 m site: a source at height $H = 3$ m releasing Q kg/h into wind u under Pasquill–Gifford stability class $s \in \{A, \dots, F\}$ produces, at a $z_R = 2$ m sensor, an enhancement

$$C(x) = \frac{Q}{2\pi U \sigma_y(x') \sigma_z(x')} \exp\left(-\frac{y'^2}{2\sigma_y^2}\right) \left[e^{-\frac{(z_R-H)^2}{2\sigma_z^2}} + e^{-\frac{(z_R+H)^2}{2\sigma_z^2}} \right], \quad (1)$$

with (x', y') the down/crosswind coordinates and $\sigma_{y,z}$ the Briggs open-country dispersion curves for class s (ground reflection is the second exponential). The response is linear in Q and closed-form — no quadrature — so the exact-posterior machinery below applies directly. Physical gates pass: exact crosswind mirror symmetry (4×10^{-14}), numerical mass-flux conservation $\iint UC \, dy \, dz = Q$ (4×10^{-6}), and 80 ppm at 100 m downwind of a 100 kg/h release in class D (literature-plausible).

Scenarios. Per scenario (Table 1, Figure 6): source location uniform on the site, $Q \sim \text{LogU}(10, 500)$ kg/h, wind 1–8 m/s in a uniform direction, stability class uniform over A–F, $N \sim \text{U}\{4, \dots, 12\}$ sensor masts at uniform positions, noise $\sigma \sim \text{LogU}(0.2, 3)$ ppm, and Bernoulli reading dropout. Each mast reports 30 time-averaged readings. Splits (10k/1k/2k/2k train/val/calib/test) are by complete scenario; a leakage audit confirms zero overlap.

Exact posteriors. For candidate cell c of a 64×64 grid with unit-rate response g_c from Eq. (1), the Gaussian likelihood $p(\mathbf{y} | c, Q) = \mathcal{N}(\mathbf{y}; Q\mathbf{g}_c, \sigma^2\mathbf{I})$ is marginalized over a LogUniform rate prior and normalized over cells to give $p(c | \mathbf{y})$ — exact for this simulator and grid. Marginalization uses a peak-aware composite quadrature (the naive fixed rule errs by many nats when the rate-likelihood peak is narrow); normalization holds to 10^{-13} , and the posterior mode recovers the true cell as $\sigma \rightarrow 0$ on identifiable scenarios (18/24; the rest are exact along-wind ridge degeneracies). Posteriors are computed for all splits: train/val become teachers (Sec. 3), calib/test the audit reference (Sec. 4).

Table 1: CH4-500 scenario prior.

Source	$\mathbf{x}_s$	U(site)	Sensor masts	N	U{4..12}
Release rate	Q	LogU(10, 500) kg/h	Mast positions	$\mathbf{x}_i$	U(site)
Wind speed	U	LogU(1, 8) m/s	Noise	σ	LogU(0.2, 3) ppm
Wind direction	θ	U(0, 2π)	Reading dropout		U(0, 0.2)
Stability class	s	U{A..F}	Release height	H	3 m (fixed)

3 Method

Localizer (fixed across all models). A permutation-invariant DeepSets network (4.87M params): each retained reading is tokenized as $[\text{Fourier}(x_i), \text{Fourier}(y_i), \text{Fourier}(t_k), \text{asinh}(y_{ik}/\sigma)]$, embedded by a shared MLP, pooled by masked mean and max, concatenated with a meteorology-context MLP over $(\mathbf{u}, s, \sigma, N)$, and decoded to a softmax over the grid. Inference is 0.48 ms/scenario. Training uses AdamW, cosine schedule, batch 256, early stopping. Each batch is augmented by a random D4 symmetry of the square applied jointly to sensors, wind, and source — an exact symmetry under which the prior is invariant; without it this network memorizes the training set with zero generalization (Figure 12).

M0: point-label supervision. The standard target: cross-entropy to a smoothed one-hot bump at the realized source. This teaches location, not uncertainty shape.

M1: physics-posterior distillation. We replace the target with the scenario’s exact posterior (Eq. (1) $\rightarrow$ likelihood $\rightarrow$ posterior), tempered by an isotropic Gaussian of 0.75 cells,

$$\mathcal{L}(\theta) = - \sum_c [G_{0.75} * p(\cdot | \mathbf{y})](c) \log p_\theta(c | \mathbf{y}), \quad (2)$$

with the teacher grid permuted by the same D4 element as the inputs, and early stopping on validation teacher cross-entropy. Everything else — architecture, data, optimizer, inference cost — is identical to M0. Tempering is necessary: with the raw teacher, identifiable-scenario inefficiency is $11.3\times$ vs. M1’s $8.5\times$ and training is unstable (Figure 8); near-delta targets give gradients dominated by cells the student cannot yet match.

Conformal regionization (the measurement control). Every model, and the exact posterior itself, is wrapped by the identical randomized highest-density-mass split conformal procedure: a nonconformity score on 2000 calibration scenarios, a threshold at the $\lceil (n+1)(1-\alpha) \rceil$ -th value, regions filled by descending probability. Under exchangeability each wrapped model has $\Pr(c^* \in C_\alpha) \geq 1 - \alpha$. This equalizes coverage across models, so region-size differences isolate sharpness. (One numerical note: the standard mass-above score saturates in floating point on sharp maps and silently loses coverage; we compute in complement space, which is exact-arithmetic-equivalent.)

Audit metrics. With learned and exact regions both at verified 90% coverage, the per-scenario area ratio (learned/exact) is the *inefficiency factor*; 1 means the localizer extracts all the information the readings contain. A dose–response protocol adds/removes sensors, contrasts geometries, and sweeps noise on constructed sites, and reports learned-vs-exact area-response slopes. Because the plume is thin and steady, many random sparse layouts are weakly identifiable, so we report inefficiency stratified by oracle difficulty rather than a single overall median.

4 Results

4.1 Point-label supervision is calibrated but not actionable

Conformal M0 attains near-nominal coverage (0.90, verified even on top of the discretized oracle; Figure 13) — and draws the same over-broad region almost regardless of how much the readings actually constrain the source (Figure 1, middle row). On identifiable scenarios (oracle region $< 10\%$ of the site) its regions are a median $32.5\times$ larger than the exact posterior permits, and its area barely responds to noise (slope ratio 0.001). Point accuracy is unremarkable but fine (median MAP error

Table 2: CH4-500 results (test split, $n = 2000$; seed 1; three-seed spread in Table 3). Inefficiency = learned/oracle region area at matched 90% coverage, reported on identifiable scenarios (oracle region < 10% of site).

	M0 (point labels)	M1 (distilled)	Exact oracle	Peak sensor
Coverage @ 90%	0.900	0.912	0.894	—
Inefficiency, identifiable (median)	$32.5\times$	$8.5\times$	$1\times$	—
Noise-response slope ratio	0.001	0.72	1.0	—
JSD to oracle (nats)	0.377	0.235	0	—
MAP error (median, m)	200	162	164	205
Inference latency (ms)	0.48	0.48	—	—

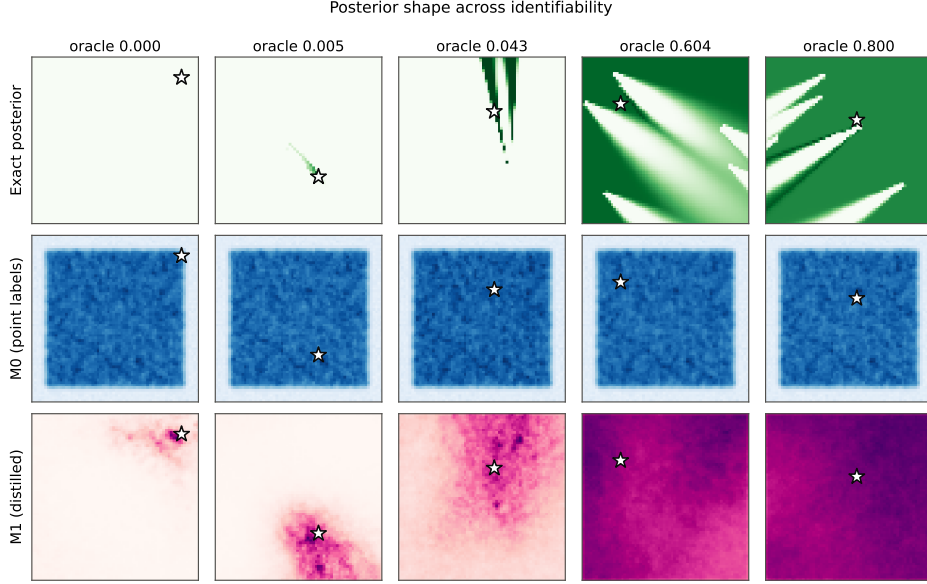


Figure 1: **Posterior shape across identifiability.** Columns: five test scenarios ordered by oracle HPD-90 area. Rows: exact posterior, M0, M1 (star = true source). The exact posterior collapses to sharp along-wind ridges when identifiable; M0 draws a similar broad blob regardless; M1 tracks the physical shape.

162–200 m, near the oracle’s 164 m and better than the 205 m peak-sensor heuristic): what point labels fail to teach is not *where*, but *how unsure* (Figure 2).

4.2 Physics-posterior distillation closes most of the gap

Changing only the training target (M1) cuts identifiable-scenario inefficiency from $32.5\times$ to $8.5\times$ (bootstrap 95% CIs [28.7, 38.2] vs. [7.5, 11.1]; sharper on 94% of scenarios; paired Wilcoxon $p < 10^{-12}$), lowers JSD to the exact posterior from 0.377 to 0.235, and improves median MAP error slightly (200 $\rightarrow$ 162 m) — all at unchanged inference cost (Table 2, Figure 2). The improvement is concentrated exactly where the data are informative (Figure 1) and is seed-stable with no metric overlap (Table 3). The raw-teacher failure and the memorization-without-augmentation collapse (Figure 12) are the two ingredients that make the physics signal usable.

4.3 Dose–response: distillation restores physically correct behavior

On constructed sites we vary information and measure how region area responds (Figure 3, 11). **Sensor masts:** M1’s area contracts as masts are added at $0.75\times$ the oracle rate with rank agreement 0.92, while M0 scarcely responds (slope ratio near zero, negative rank agreement). **Noise:** M1 tracks the oracle’s noise response (slope ratio 0.72, agreement 0.71) where M0 is flat (0.001). **Geometry:**

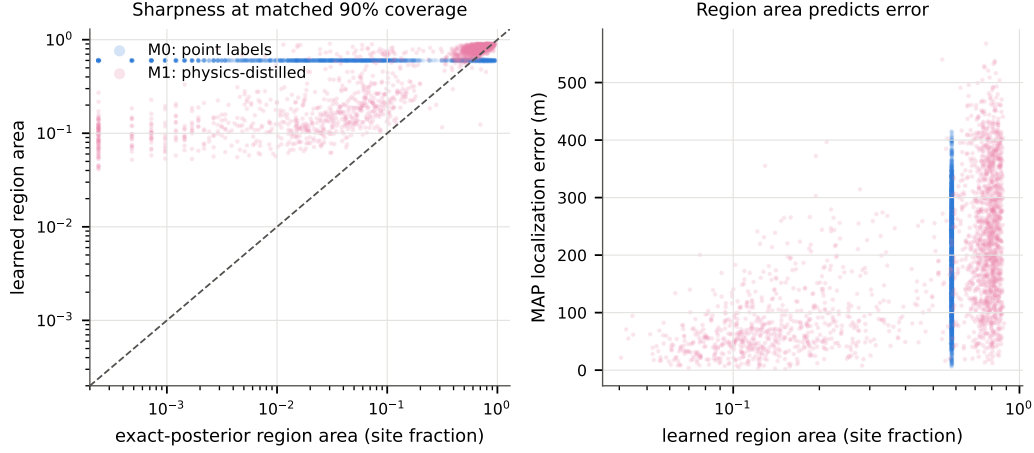


Figure 2: **The amortization gap, measured.** Left: learned vs. exact region area at matched 90% coverage; distillation moves the identifiable-scenario cloud toward the diagonal. Right: region area predicts localization error (m), supporting area as an abstention signal.

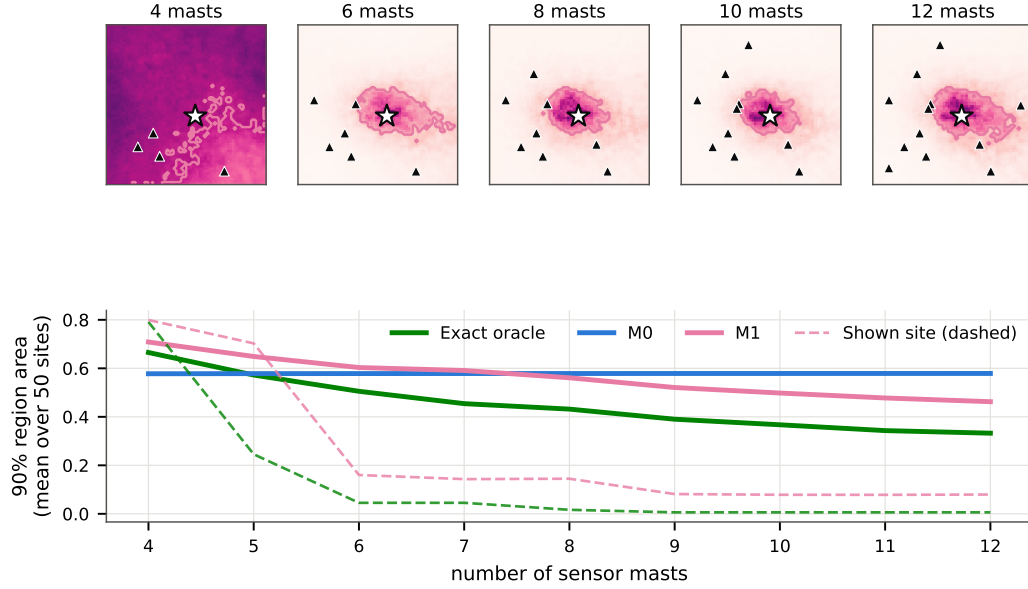


Figure 3: **Marginal value of a sensor mast.** Top: one site as masts accrue 4 $\rightarrow$ 12 (M1 heatmap; region as fill; star = source). Bottom: mean 90% region area vs. mast count for oracle, M0, M1 over 50 sites.

M1 agrees with the oracle on which confined layouts widen the region (agreement 0.86 vs. M0's 0.27). In each case the point-label model reacts to information far more weakly than the physics dictates, and the physics target largely repairs it.

4.4 Where the gap lives, and a network-design finding

Stratifying the audit (Figure 4) shows the gap is confined to information-rich scenarios: with sparse masts (4–6) or in weakly identifiable geometries both models sit near the physical limit ($\approx 1\times$) — the sensor network, not the model, binds — while the $32\times \rightarrow 8.5\times$ improvement is entirely in the dense, identifiable regime. Two consequences. First, an investment rule: below ~ 7 masts, add sensors; above, improve the model. Second, and more striking: *over half* (57%) of scenarios with randomly

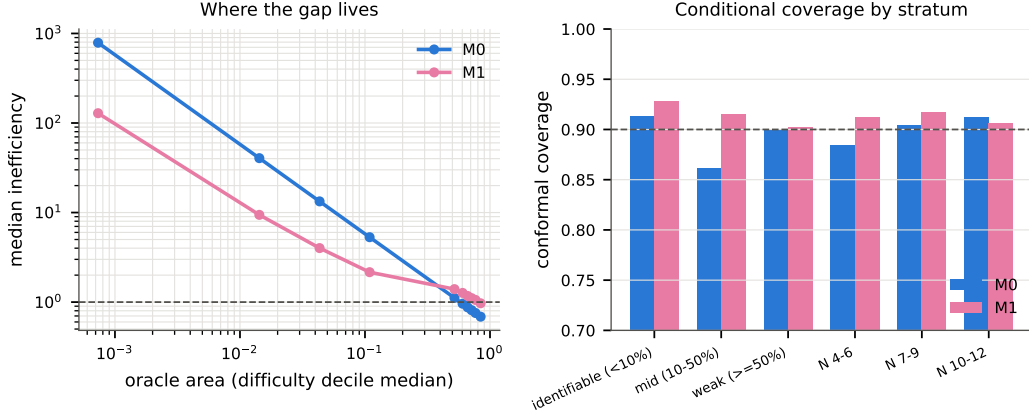


Figure 4: **Stratified audit.** Left: median inefficiency by oracle-difficulty decile — the gap lives where the oracle is sharp. Right: conditional coverage by stratum; distillation is sharper and more uniformly calibrated.

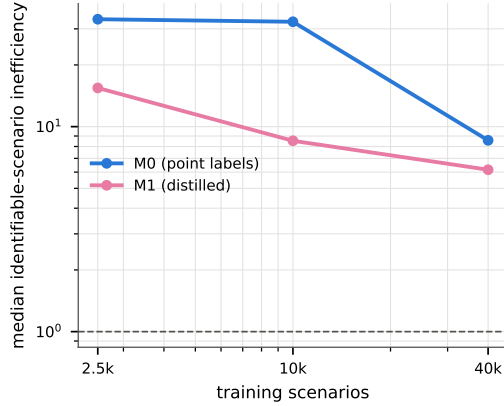


Figure 5: **Data-scaling.** Identifiable-scenario inefficiency vs. training size; no plateau by 40k, distillation offset ahead by roughly $4\times$ in data.

placed masts are weakly identifiable (oracle region $\geq 50\%$ of the site), because thin steady plumes simply miss randomly placed sensors — a quantitative argument that methane monitoring networks require deliberate placement, not density alone. Distillation also makes conditional coverage more uniform across strata (0.906–0.928 vs. M0’s 0.862–0.913).

4.5 The gap is data-limited; distillation is worth extra data

Retraining at $\{2.5, 10, 40\}k$ scenarios (Figure 5) shows identifiable-scenario inefficiency falling with data for both supervisions and *not* plateauing by 40k — so at this architecture the residual gap is predominantly a training budget limit, not a capacity floor. Distillation stays ahead throughout (M1@10k $\approx$ M0@40k: $8.5\times$ vs. $8.6\times$), i.e. the physics target buys roughly the sharpness of a $4\times$ larger point-labeled set at this range — a direct data-efficiency statement for standing up attribution models on new facility types.

5 Discussion

Operational reading. At facility scale the median identifiable-scenario region shrinks under distillation while coverage holds, and region area is a direct proxy for ground-survey effort, so sharper regions mean fewer candidate sites per satellite alert and faster repair; region area also predicts

Table 3: Seed stability (three seeds; mean [min, max]).

	M0	M1
Coverage @ 90%	0.905 [0.900, 0.909]	0.906 [0.898, 0.912]
Inefficiency, identifiable (median)	$32.6\times$ [32.5, 32.7]	$8.6\times$ [8.5, 8.7]

localization error (Figure 2), supporting an abstention rule (“flag, don’t dispatch, when the region is large”). The audit doubles as procurement infrastructure: a regulator can screen any vendor localizer for calibrated *and* sharp attribution against exact references. These are readings of a controlled testbed, offered as design guidance, not deployment claims.

Relation to prior work. Amortized neural posterior estimation and conformal UQ are standard [Angelopoulos and Bates, 2021; Cabezas et al., 2025]; Distill-Belief [Shi et al., 2026] distills a Bayes-correct particle-filter teacher for source localization. We distill *exact* grid posteriors, audit sharpness against the same posteriors at matched coverage, and use conformal prediction as a measurement control rather than a contribution. Gaussian-plume inversion is the standard methane STE model [Newman et al., 2024]; we add the amortized-uncertainty audit and the physics-derived training signal.

Limitations. CH4-500 is idealized: steady-state 2-D-sensed plume, flat terrain, known stability class, fixed and known release height, Gaussian noise, no background or obstacles; “exact” is exact for this simulator, not the real atmosphere. Coverage is marginal and in-distribution. Distillation requires a tractable per-scenario posterior at training time (automatic here; an approximation problem in richer simulators). We study one architecture. The path to deployment runs through controlled-release recalibration and integration with real monitoring networks — future work, not claimed here.

6 Conclusion

Trustworthy methane attribution, not detection, gates methane response. Using the physics simulator behind an attribution tool to compute exact per-scenario posteriors, we show those posteriors can both audit a learned localizer against the physical information limit and, as a training signal, close most of the gap between a calibrated-but-vague baseline and that limit — cutting identifiable-scenario region inefficiency from $32\times$ to $8.5\times$ at no inference cost, with a stratified audit that tells monitoring programs whether to invest in sensors or software. The methane-configured benchmark and its exact posteriors are released.

Reproducibility

One config drives every stage; scenario regeneration is bit-identical (SHA-256 verified). We release the plume simulator, frozen scenarios, exact per-scenario posteriors for all splits, dose-response sets, checkpoints, and the audit pipeline. Total compute: ≈ 3 GPU-hours on one L40S.

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A Benchmark Gallery and Additional Figures

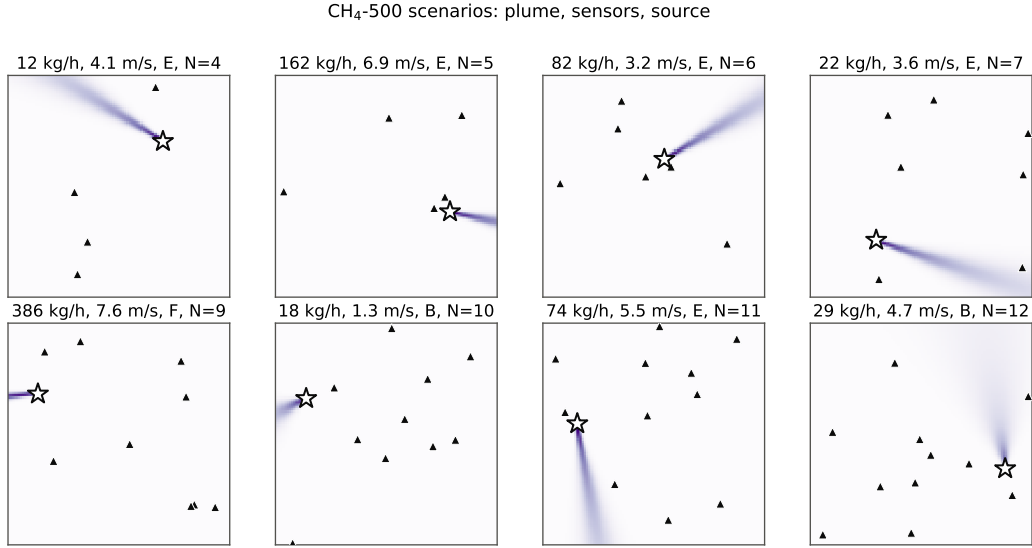


Figure 6: **CH₄-500 gallery**. Eight test scenarios: steady plume (purple), sensor masts (triangles), true source (star), with release rate, wind, stability class, mast count.

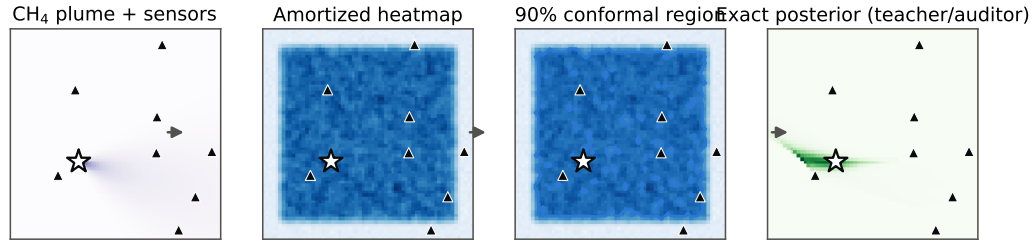


Figure 7: **Pipeline**. Plume + sensors → amortized heatmap → 90% conformal region; the exact posterior is teacher (train/val) and auditor (calib/test).

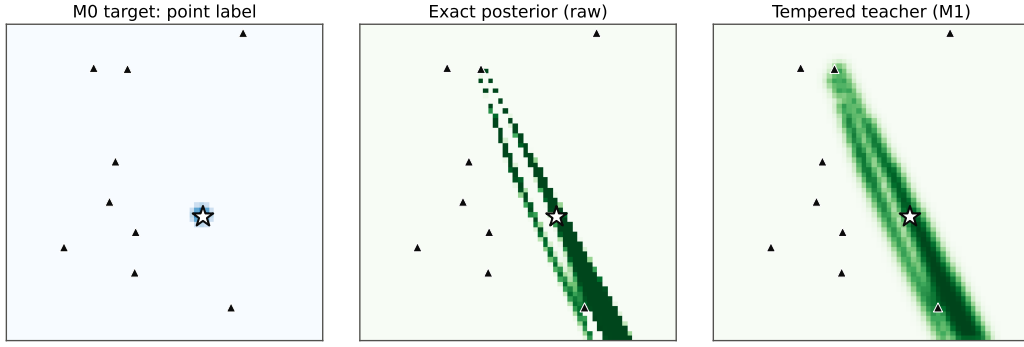


Figure 8: **Three supervision signals** for one scenario: point-label bump (M0), raw exact posterior (optimization trap), tempered teacher (M1).

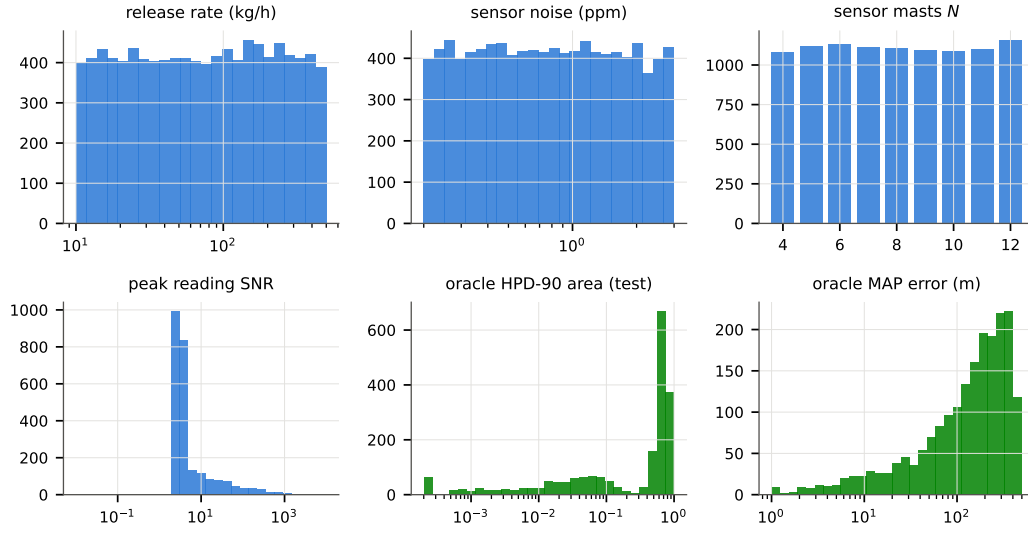


Figure 9: **Benchmark statistics.** Prior draws (top); reading SNR, oracle HPD-90 area, oracle MAP error in meters (bottom).

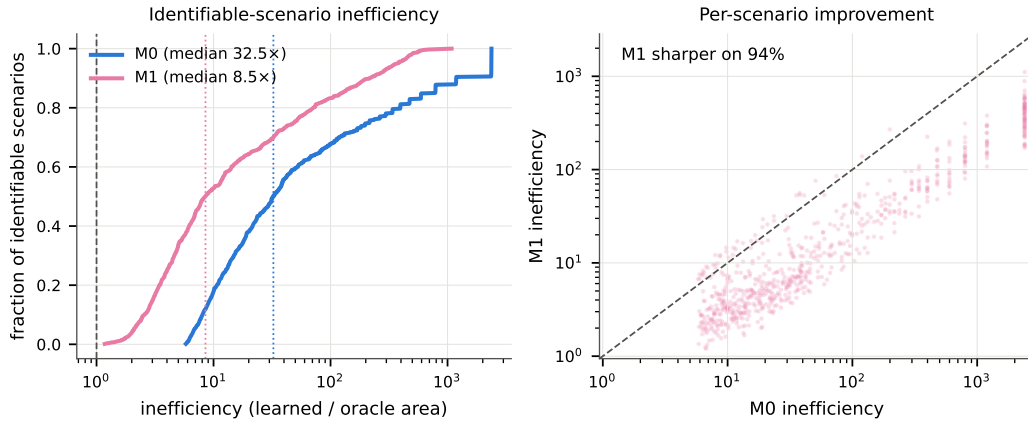


Figure 10: **Per-scenario improvement** (identifiable subset): inefficiency ECDFs and paired ratios.

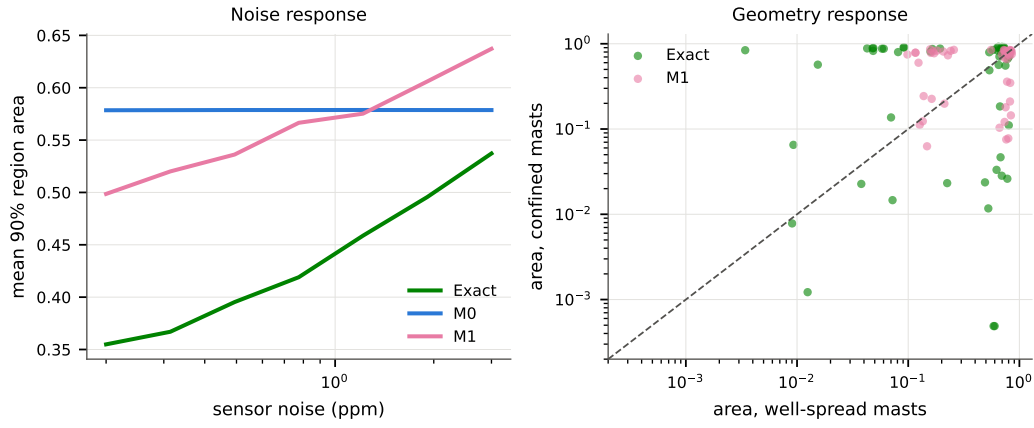


Figure 11: **Noise and geometry knobs.** Left: region area vs. sensor noise (ppm). Right: matched geometry pairs; points above the diagonal widen under confinement.

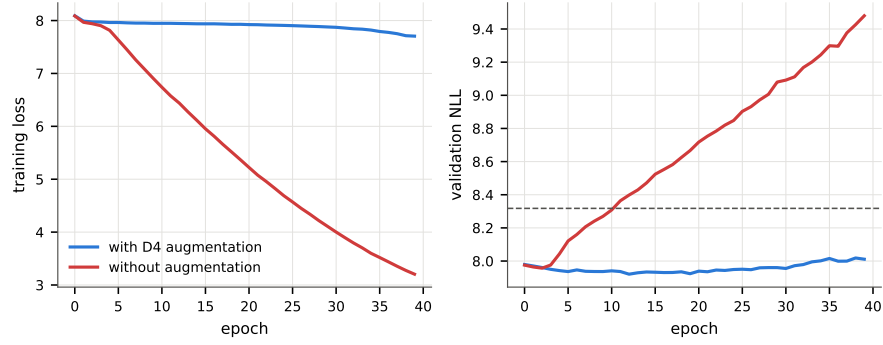


Figure 12: **Memorization without symmetry augmentation.** Training loss falls either way; validation NLL only leaves the uniform level with D4 augmentation.

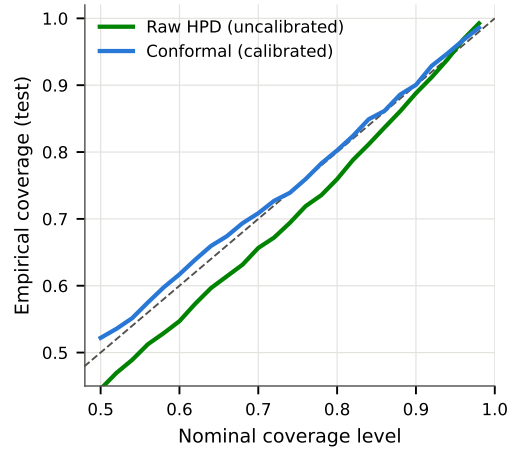


Figure 13: **Coverage vs. nominal (M0):** conformal tracks the diagonal, raw HPD undercovers; coverage is thereby equalized across models.

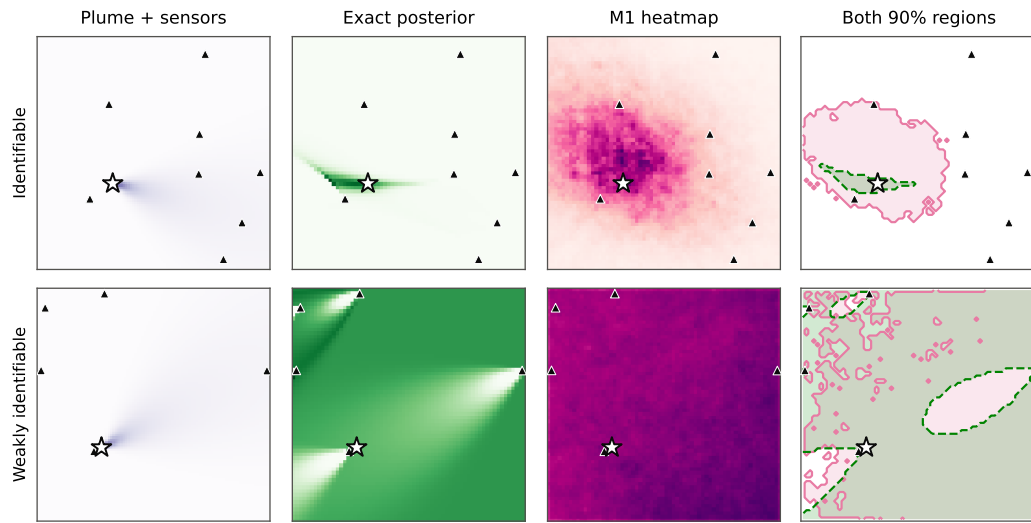


Figure 14: **Identifiable vs. weakly identifiable** scenario: plume, exact posterior, M1 heatmap, both 90% regions (solid = M1, dashed = exact).